

Acids and oxides

When an acid reacts with an oxide (a compound with oxygen), it forms a salt and water. The hydrogen ions from the acid and the oxide ions form the water molecules. The cation (the positively charged portion of the oxide)—generally a metal ion—then forms a salt with the anion (the negative part of the acid).

▽ General equation

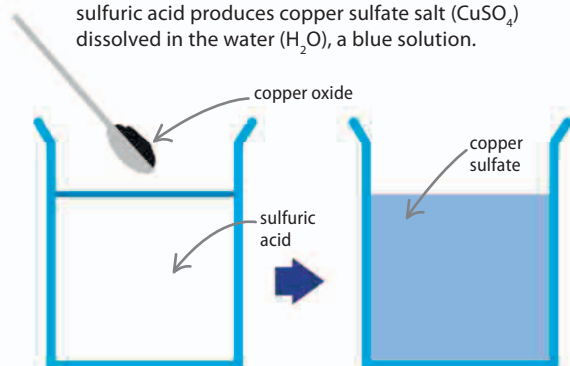
The acid–oxide reaction has the same products as an acid–base reaction (see page 144).

acid + oxide = salt + water



▽ Sulfuric acid (H_2SO_4) plus copper oxide (CuO)

The black copper oxide powder added to colorless sulfuric acid produces copper sulfate salt (CuSO_4) dissolved in the water (H_2O), a blue solution.



Acids and carbonates

When an acid reacts with a carbonate, the products are a salt, water, and carbon dioxide. A carbonate is an ionic compound in which the anion is CO_3^{2-} . The carbonate ion is displaced in the reaction by the anion from the acid. The carbonate ion then reacts with the free hydrogen ion to form water and carbon dioxide.

▽ General equation

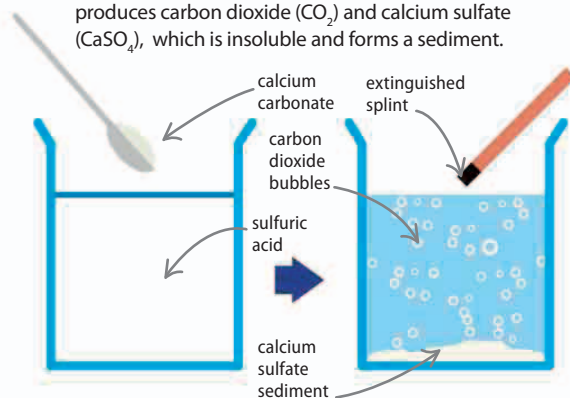
The acid–carbonate reaction produces a salt, water, and also carbon dioxide gas.

acid + carbonate = salt + carbon dioxide + water



▽ Sulfuric acid (H_2SO_4) + calcium carbonate (CaCO_3)

White calcium carbonate powder added to sulfuric acid produces carbon dioxide (CO_2) and calcium sulfate (CaSO_4), which is insoluble and forms a sediment.



Acids and sulfites

A sulfite is a compound made up of at least one cation, often a metal, and a sulfite SO_3^{2-} ion. When a sulfite reacts with an acid, the products are a salt, water, and sulfur dioxide. The sulfite ion is displaced in the reaction by the anion from the acid. The sulfite ion then reacts with the free hydrogen to form water and sulfur dioxide gas.

▽ General equation

This reaction is very similar to that of the acid–carbonate reaction, except sulfur dioxide gas is formed instead of carbon dioxide.

acid + sulfite = salt + sulfur dioxide + water



▽ Hydrochloric acid (HCl) plus copper sulfite (CuSO_3)

Blue copper sulfite crystals added to clear hydrochloric acid produces the green salt copper chloride (CuCl_2), dissolved in water, and smelly sulfur dioxide (SO_2).

